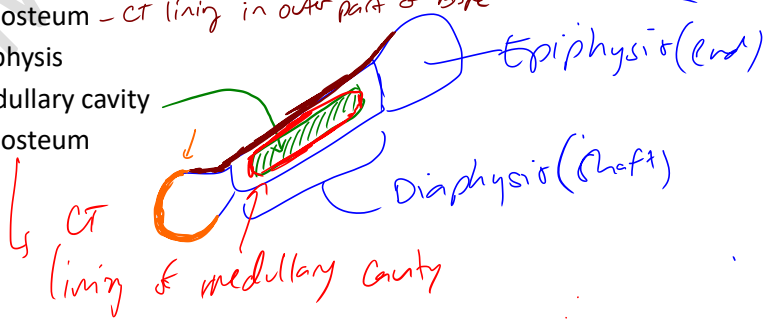
DMF DENTAL TRAINING CENTER: FINAL COACHING

General Physiology

By: Mickhail Q. Francisco, DMD

1. A patient presents with excessive thirst and excessive urination. Which condition is most characteristically associated with these findings?
   A. Addison disease
   B. Diabetes insipidus
   C. Cushing syndrome
   D. Hyperthyroidism
2. Which structure stabilizes a nasotracheal tube during airway management?
   A. Tourniquet
   B. Cervical collar
   C. Adhesive tape
   D. Facemask
3. Which lobe of the brain is primarily responsible for hearing?
   A. Frontal lobe
   B. Parietal lobe
   C. Temporal lobe
   D. Occipital lobe
4. Ejection of blood by the heart per minute is termed:
   A. Stroke volume
   B. Cardiac output
   C. Heart rate
   ~~D. Pulse pressure~~
5. The main function of the central nervous system is to:
   A. Synthesize, Produce and Secrete hormones
   B. Receive, integrate, and respond to stimuli
   C. Control the function of the organs through somatic nervous system
   D. Store neurotransmitters
6. The cavity located at the center of the diaphysis of a long bone is the:
   A. Periosteum
   B. Epiphysis
   C. Medullary cavity
   D. Endosteum

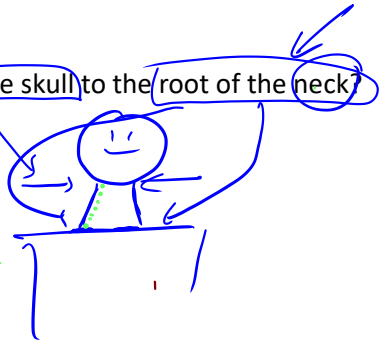


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7. Most amino acids filtered by the kidney are reabsorbed in the:
   A. Distal convoluted tubule
   B. Loop of Henle
   C. Collecting duct
   D. Proximal convoluted tubule

8. The lymph node located superficial to the buccinator muscle is the:
   A. Deep cervical node
   B. Buccal node
   C. Jugulodigastric node
   D. Submental node

9. Which lymphatic structure extends from the base of the skull to the root of the neck?
   A. Submental node
   B. Buccal node
   C. Deep cervical node
   D. Parotid node

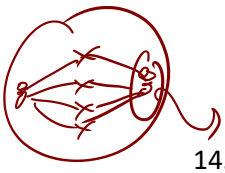


10. The thoracic duct drains into the junction of the:
   A. Right subclavian and right internal jugular veins
   B. Left subclavian and left internal jugular veins
   C. Left external jugular and left subclavian veins
   D. Right external jugular and right subclavian veins

11. Study of the masticatory system and its disorders is called:
   A. Orthodontics
   B. Deglutition
   C. Gnathology
   D. Prosthodontics

12. Different color of the two iris is termed:
   A. Presbyopia
   B. Diplopia
   C. Dichromacy
   D. Heterochromia

13. The primary site of protein synthesis in the cell is the:
   A. Lysosome
   B. Ribosome
   C. Golgi apparatus
   D. Centriole



14. Programmed cell death is known as:
   A. Necrosis
   B. Autolysis

C. Apoptosis - physiologic (inflammation)
D. Degeneration - retrogressive change / deterioration of cell (reversible/irreversible)

15. Injury to the occipital lobe is most likely to affect:
    A. Hearing
    B. Smell
    C. Vision
    D. Balance
16. Fertilization normally occurs in the:
    A. Uterus
    B. Cervix
    C. Ampulla of the uterine tube
    D. Ovary



17. The myelin sheath in the peripheral nervous system is produced by:
    A. Astrocytes - Blood-Brain Barrier
    B. Oligodendrocytes - CNS myelin sheath
    C. Schwann cells
    D. Microglia - Phagocyte
18. Which substance is considered a major nonprotein nitrogen compound in blood?
    A. Glucose - $C_6H_{12}O_6$
    B. Urea
    C. Albumin X
    D. Fibrinogen X

-in, -ogen, -ase - protein, protein enzyme

19. The attachment of a muscle to a bone is called a:
    A. Ligament - Bone to Bone
    B. Bursa - cushion
    C. Tendon
    D. Fascia - covering of organ (muscle)
20. Synovial fluid primarily functions to:
    A. Form bone
    B. Lubricate joints
    C. Produce blood cells
    D. Store minerals
21. Which ion is most responsible for maintaining the resting membrane potential?
    A. Calcium
    B. Potassium
    C. Magnesium
    D. Iron

Na^+ or K^+

22. Local anesthetics primarily block:
    A. Potassium channels
    B. Calcium channels
    C. Chloride channels
    D. Sodium channels
23. A bedridden patient is at greatest risk of developing:
    A. Cellulitis
    B. Deep vein thrombosis
    C. Osteoporosis
    D. Migraine
24. Tachycardia with cold extremities is most likely due to:
    A. Parasympathetic stimulation
    B. Sympathetic stimulation
    C. Vestibular stimulation
    D. Cortical inhibition
25. Crossing over occurs during:
    A. Mitosis
    B. Cytokinesis
    C. Meiosis I
    D. Meiosis II
26. The normal platelet count is important in evaluating:
    A. Bleeding tendency
    B. Vision
    C. Respiration
    D. Digestion
27. Bodily side-to-side movement of the mandible is called:
    A. Retrusion
    B. Protrusion
    C. Lateral excursion
    D. Depression
28. In a patient with orthopnea, which dental position should generally be avoided?
    A. Upright position
    B. Semi-supine position
    C. Supine position
    D. Sitting position
29. Dental treatment for a dialysis patient is preferably performed:
    A. During dialysis
    B. Immediately before dialysis

C. On the day after dialysis
D. During heparin infusion

30. Ventricular contraction normally causes:
A. Opening of AV valves
B. Closure of AV valves
C. Closure of aortic valve before ejection
D. Opening of pulmonary veins

31. Which event occurs immediately after atrial systole?
A. Ventricular filling ends and isovolumetric contraction begins
B. Semilunar valves open
C. Ventricular ejection begins
D. Atrial relaxation ends

32. A patient taking heparin experiences prolonged bleeding after a minor cut. Which laboratory value is most likely prolonged?
A. Bleeding time only
B. Prothrombin time only
C. Partial thromboplastin time
D. Hematocrit

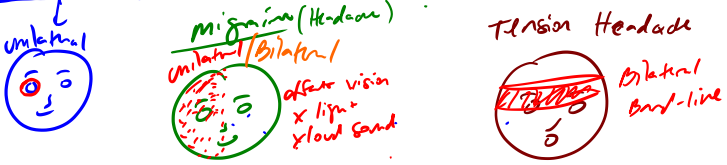
33. Which statement best differentiates acromegaly from gigantism?
A. Both occur before epiphyseal closure
B. Both occur after epiphyseal closure
C. Acromegaly occurs after epiphyseal closure, gigantism before closure
D. Acromegaly is caused by thyroid hormone excess

34. The action of the lateral pterygoid muscle includes:
A. Elevation of the mandible
B. Retraction of the mandible
C. Protrusion of the mandible
D. Fixation of the hyoid bone

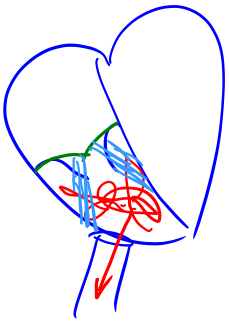
35. Distention of the minor calyces is most likely associated with:
A. Urinary obstruction
B. Hyperventilation
C. Gastritis
D. Hepatitis

36. Which vessel receives lymph from most of the body?
A. Right lymphatic duct
B. Thoracic duct
C. Cisterna chyli only
D. Azygos vein

37. A deficiency of antidiuretic hormone produces:
    A. Diabetes mellitus
    B. Diabetes insipidus
    C. Hyperthyroidism
    D. Acromegaly
38. Closure of the semilunar valves produces which heart sound component?
    A. S1 – AV valve close
    B. S2
    C. S3
    D. S4
39. The principal function of red bone marrow is:
    A. Fat storage – yellow bone marrow
    B. Hematopoiesis – Blood production
    C. Calcium absorption – small intestin (duodenum)
    D. Cartilage formation – chondroblast
40. The Trendelenburg position involves:
    A. Head elevated above feet
    B. Body lying flat with feet elevated above the head
    C. Sitting upright
    D. Lateral recumbent position
41. A patient suffering from cluster headaches usually experiences pain:
    A. Bilaterally in the occiput
    B. Around one eye
    C. Over the vertex only
    D. In the TMJ region only



42. The primary physiological role of sodium during nerve impulse conduction is:
    A. Repolarization – K^+
    B. Depolarization
    C. Protein synthesis
    D. Muscle relaxation
43. Which hormone is directly responsible for increasing blood calcium levels?
    A. Calcitonin – ↓ Ca^{++} in Blood
    B. Insulin – ↓ glucose in blood
    C. Parathyroid hormone
    D. Cortisol – for stress; ↑ glucose in blood; ↑ BMR
44. Which structure conducts impulses from the AV node to the ventricles?
    A. SA node
    B. Purkinje fibers



C. Bundle of His
D. Chordae tendineae

45. The major force driving filtration in the renal glomerulus is:
A. Oncotic pressure
B. Capsular pressure
C. Hydrostatic pressure of glomerular capillaries
D. Osmotic pressure of tubules

46. Which blood vessel carries oxygenated blood from the lungs to the heart?
A. Pulmonary artery
B. Superior vena cava
C. Aorta
D. Pulmonary veins

47. During ventricular systole, which combination is correct?
A. AV valves open; semilunar valves closed
B. AV valves closed; semilunar valves open during ejection
C. Both valves closed throughout systole
D. Both valves open simultaneously

48. Which of the following best explains the rapid onset of action of local anesthetics in nerve tissue?
A. Blockade of calcium influx
B. Blockade of sodium channels preventing depolarization
C. Increased potassium permeability
D. Increased chloride entry

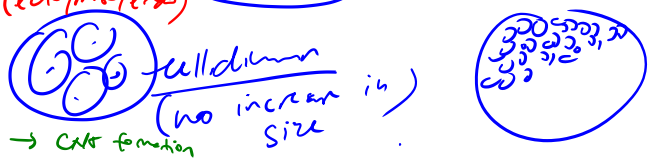
49. Which phase of the cardiac cycle has all four heart valves closed?
A. Rapid ventricular filling
B. Atrial systole
C. Isovolumetric contraction
D. Ventricular ejection

50. The principal mechanism preventing loss of filtered amino acids in urine is:
A. Passive diffusion in collecting ducts
B. Active transport in the proximal tubule
C. Filtration barrier selectivity
D. Distal tubular secretion

51. A patient with severe dehydration has markedly elevated plasma osmolarity. Which hormone secretion would increase most directly?
A. Aldosterone
B. Growth hormone

C. Antidiuretic hormone
D. Thyroxine

52. Which embryologic event immediately follows fertilization?
    A. Gastrulation
    B. Organogenesis
    C. Cleavage divisions
    D. Neurulation



53. A kind of muscle found in our body which are extremely fatigue resistant?
    a. Skeletal muscles
    b. Smooth muscles
    c. Cardiac muscles
    d. None of the above

54. Preload is equivalent to which of the following?
    a. It is equivalent to the amount of blood leaving the heart after ventricular contraction
    b. It is equivalent to the rate of venous return
    c. It is equivalent to the atrial end-diastolic volume
    d. It is equivalent to the atrial blood pressure and atrial blood volume

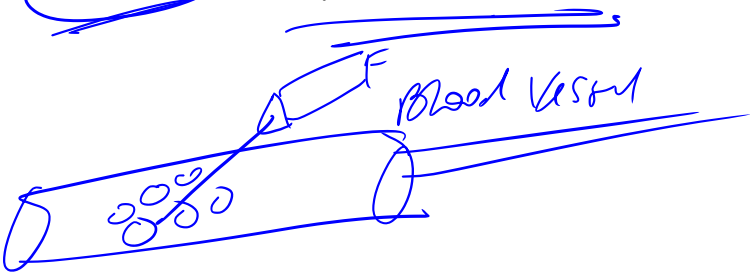
55. The neural tube starts to form around how many days after fertilization?
    a. 10 days
    b. 20 days
    c. 35 days
    d. 55 days

56. All of the following are enzymes activated by trypsin except:
    a. Prolipase
    b. Pepsinogen
    c. Chymotrypsinogen
    d. Proelastase

57. All of the following are controlled by the Vagus nerve, except one. Which is it?
    a. Blood pressure
    b. Intestinal motility
    c. Salivary flow
    d. Heart rate

58. In the process of coagulation cascade, clotting factor XII activates clotting factor IX. The activated clotting factor IX will activate clotting factor VIII.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true; the second statement is false
    d. The first statement is false; the second statement is true

59. Listed below are the possible causes of air embolism that may block the heart chamber. Which is usually not associated?
    a. IV fluid administration



b. Dialysis line
c. IM injection
d. Cardiopulmonary bypass

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61. The compression to breath ratio for infants and children when CPR is done by a single rescuer is:
    a. 30:2
    b. 15:2
    c. 30:1
    d. 15:1

62. The compression to breath ratio for infants and children when CPR is done by two rescuers is:
    a. 30:2
    b. 15:2
    c. 30:1
    d. 15:1

63. The compression to breath when doing CPR for adults is:
    a. 30:2
    b. 15:2
    c. 30:1
    d. 15:1

64. When performing one-man E-C clamp technique during CPR, the index and thumb will form a "C" to _____ then the third, fourth and fifth finger will form an "E" to _____
    a. Exert an upward pressure on the mask: apply pressure on the mandible and lift it forward.
    b. Exert a downward pressure on the mask: apply pressure on the mandible and lift it forward.
    c. Exert an upward pressure on the mask: apply pressure on the mandible and lift it backward
    d. Exert a downward pressure on the mask: apply pressure on the mandible and lift it backward.

65. At resting states, blood volume is seen highest in which of the following parts of the circulatory system?
    a. Veins and venules
    b. Arteries and arterioles
    c. Capillaries
    d. Heart
    e. Pulmonary circulation

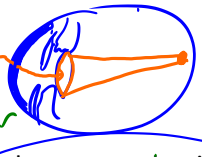


66. This is the muscle that pulls the tongue superiorly and posteriorly
    a. Genioglossus
    b. Palatoglossus
    c. Styloglossus
    d. Hyoglossus

67. Removal of this organ may significantly affect the calcium levels in the blood
    a. Thyroid gland
    b. Liver
    c. Salivary glands
    d. Gall bladder



68. The primary refracting structure of the eye is the _____
    a. Cornea
    b. Aqueous humor
    c. Lens
    d. Vitreous humor

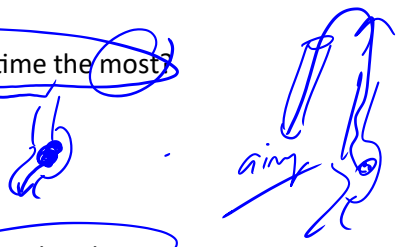


69. It is a false statement regarding saltatory conduction
    a. Myelinated regions are skipped by the electrical signal
    b. It is the propagation of signals in myelinated axons
    c. Electrical signals leap in between successive myelin sheath gaps
    d. The electrical waveform travels down the entire length of the axon

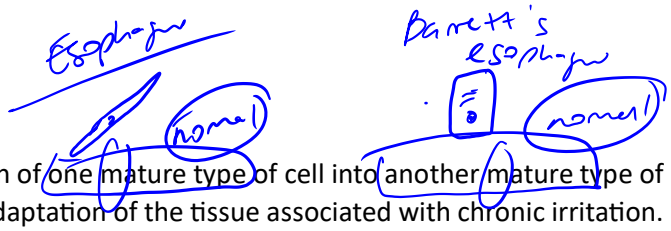
70. Hyperventilation will result to all of the following, except one. Which is it?
    a. hypocapnia
    b. ~~decreased cerebral blood flow~~
    c. increased PCO2
    d. ~~respiratory alkalosis~~

71. Dietary calcium is primarily absorbed from which of the following segments of the alimentary canal?
    a. Oral cavity
    b. Stomach
    c. Small intestine
    d. Large intestine

72. Which of the following increases the gastric emptying time the most?
    a. White rice
    b. Blended meals
    c. Liquid drinks
    d. Fats



73. This phase of the cardiac cycle happens just after aortic valve closes
    a. Isovolumetric relaxation
    b. Isovolumetric contraction
    c. Systolic ejection
    d. Diastolic filling



74. It is the transformation of one mature type of cell into another mature type of cell in a tissue. It is commonly due to adaptation of the tissue associated with chronic irritation.
    a. Metaplasia
    b. Dysplasia
    c. Anaplasia
    d. Dysgrasia

2. Stored serotonin in platelets once released will have which of the following effects?
    a. Decreased bronchial constriction
    b. Increased cardiac output
    c. Increased venous return
    d. Increased vasoconstriction of vessels

75. The most important neurotransmitter for learning is _____
    a. Acetylcholine
    b. Serotonin
    c. Glutamate
    d. Dopamine

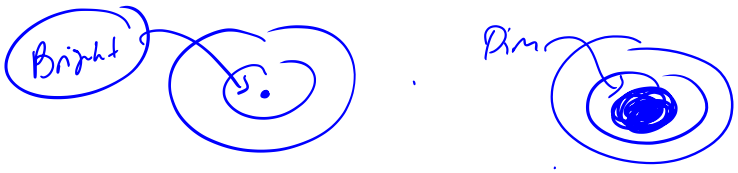
76. A type of hemoglobin found in patients with alpha-thalassemia
    a. Hemoglobin H
    b. Hemoglobin S
    c. Hemoglobin M
    d. Hemoglobin A
    e. Hemoglobin F

77. Excretion of calcium is primarily facilitated by _____
    a. GI tract
    b. Kidney
    c. Lungs
    d. Liver

78. During muscular contraction, ATP is the immediate source of energy. However, the ATP pool is extremely small that is why it must be replenish continuously during muscular function. Which of the following is not a source for new ATP?
    a. Creatine phosphate
    b. Glycogen
    c. Cellular respiration
    d. Lactic acid

79. Renin-Angiotensin-Aldosterone System regulates blood volume and electrolytes (especially Sodium) in the blood. Which of the following is not a stimulus for its activation?
    a. Sodium deficiency
    b. Blood pressure lower than the average BP of the patient
    c. Decrease blood volume
    d. A sudden increase in the blood pressure

80. In a man at rest, the amount of blood which passes through the capillaries of the systemic circulation per minute is the equal to _____
    a. The amount of blood passing through the aorta per minute
    b. Half the amount of blood passing through the aorta per minute
    c. 1/3 of the amount of blood passing through the aorta per minute
    d. ¼ the amount of blood passing through the aorta per minute
81. The low serum phosphate level in hyperparathyroidism is due to which of the following?
    a. Decrease phosphate absorption in the GI tract
    b. Increase tubular secretion of phosphate in the renal tubules
    c. Increase formation of calcium phosphate deposits in bone
    d. Increase phosphate absorption in the GI tract
82. The following are directly involved in the process of blood clot formation except one, which is the exception?
    a. Fe2+
    b. Prothrombin – CF II
    c. Fibrinogen – CF I
    d. Stuart – CF X
83. Which of the following is not true about hematocrit
    a. Hematocrit is the ratio of the total volume of RBCs to the total blood volume.
    b. It is also known as Packed Cell Volume
    c. The average hematocrit in males is around 47% while for the females it is around 42%
    d. A high hematocrit count may indicate anemia
84. A reflex normally seen in infants that happens when the baby's head is turned to one side. It results to the movement of the arm on that side to stretch out and the opposite arm to bend up at the elbow
    a. Babinski reflex
    b. Fencing reflex
    c. Moro reflex – startle/feels like falling
    d. Rooting reflex
85. The internal body temperature is regulated by which of the following structures of the CNS?
    a. Frontal lobe
    b. Midbrain
    c. Medulla oblongata
    d. Hypothalamus
86. Which of the following is not a true statement?
    a. Serum of blood is the plasma fluid without fibrinogen
    b. Plasma fluid is blood without the formed elements
    c. Formed elements of blood includes all of the blood cells and the plasma
    d. None of the above
87. What is the reason for the temporary blindness experienced when a person enters a dimly lit room from a brightly lit room?



a. The iris dilates the lens, allowing less light to enter the eye
b. The iris constricts the lens, allowing less light to enter the eye
c. The iris initially remains constricted, allowing less light to enter the eye until the pupil gradually dilates
d. The iris constricts the pupil, allowing more light to enter the eye

88. Category of an exocrine gland where its secretion is released through sectioning of the cell including part of the cytoplasm
    a. Apocrine
    b. Accine
    c. Merocrine
    d. Holocrine

89. In the general circulation, blood pressure is lowest in Vena cava, and highest in the Aorta.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true; the second statement is false
    d. The first statement is false; the second statement is true

90. Oxyhemoglobin dissociation curve will shift to the left by the following, except one:
    a. Decrease pH
    b. Decrease CO2
    c. Decrease temperature
    d. Decrease hydrogen ion

91. Dissociation of oxyhemoglobin is due to the following, except:
    a. Activity of carbonic anhydrase
    b. Increase oxygen in the tissues
    c. Increase temperature
    d. Decrease oxygen in the tissues

92. Carbon monoxide is a known toxic molecule in the human body. Its toxic effects are primarily due to?
    a. Decreasing the respiratory rate leading to hypoxia
    b. Inhibits the dissociation of oxyhemoglobin
    c. Combines with hemoglobin to form an abnormal methemoglobin
    d. Competes with Oxygen in hemoglobin

93. These vessels are the most critical in controlling the blood flow to the peripheral tissues
    a. Large arteries
    b. Arterioles
    c. Large veins
    d. Venules

94. When a person loss his ability to smell, this part of the brain is damaged
    a. Frontal lobe
    b. Parietal lobe
    c. Occipital lobe
    d. Temporal lobe

95. These are the contractile or functional units of muscle cells
    a. Z-discs
    b. Sarcomeres
    c. Sarcoplasm
    d. Sarcolemma
96. The principal ECG or EKG intervals are P, QRS and T waves; the QRS complex represents atrial depolarization.
    a. Both statements are true
    b. Both statements are false
    c. The first statement is true; the second statement is false
    d. The first statement is false; the second statement is true
97. Which of the following represents the end of ventricular depolarization and the beginning of ventricular repolarization?
    a. S-T segment
    b. P-R interval
    c. Q-T Interval
98. This represents the length of time between depolarization of the atria and depolarization of the ventricles
    a. S-T segment
    b. P-R interval
    c. Q-T Interval
99. It represents the time it takes for the ventricles of the heart to depolarize and repolarize
    a. S-T segment
    b. P-R interval
    c. Q-T Interval
100. _____ is the neurotransmitter stimulated by nitrous oxide to be released in order to inhibit pain signaling
    a. Dopamine
    b. Serotonin
    c. Norepinephrine
    d. Glutamate